

$$4.1) p_X(x) = \frac{8}{36} x = \frac{2}{9} x \quad x = 0, \frac{1}{8}, \frac{2}{8}, \dots, 1$$

$$a) Y = 2X - 1$$

$$P(Y=y) = P(2X-1=y) = P(2X=y+1) = P(X=\frac{y+1}{2}) \\ = p_X(\frac{y+1}{2}) = \frac{2}{9} \left(\frac{y+1}{2}\right) = \frac{y+1}{9} = p_Y(y)$$

$$\text{para } y = -1; -\frac{3}{4}; -\frac{1}{2}; -\frac{1}{4}; 0; \frac{1}{4}; \frac{1}{2}; \frac{3}{4}; 1$$

$$b) Y = 128X^2$$

$$p_Y(y) = P(Y=y) = P(128X^2=y) = P(X^2=\frac{y}{128}) = \\ = P(X=\sqrt{\frac{y}{128}}) = p_X(\sqrt{\frac{y}{128}}) = \frac{2}{9} \sqrt{\frac{y}{128}}$$

$$\text{para } y = 0, 2, 8, 18, 32, 50, 72, 98, 128$$

$$c) Y = -64X^2 + 64X + 2$$

$$p_Y(y) = P(Y=y) = P(-64X^2 + 64X + 2 = y)$$

$$* p_Y(2) = p_X(0) + p_X(1) \quad * p_Y(9) = p_X(1/8) + p_X(7/8)$$

$$* p_Y(14) = p_X(2/8) + p_X(6/8)$$

$$* p_Y(17) = p_X(3/8) + p_X(5/8)$$

$$* p_Y(18) = p_X(1/2)$$

$$y = 2, 9, 14, 17, 18, 17, 14, 9, 2$$

$$\Rightarrow f_X(x) = \frac{1}{\pi(1+x^2)} \quad x \in \mathbb{R}$$

b) SEBASTIAN
KOHN

d) $Y = 64X^2 - 96X + 128$

idem inciso anterior

c) 4.2) $Y = |\sin(\frac{1}{2}\pi X)|$ $X \sim Po(2)$

y toma valores 0 y 1 $\rightarrow$ amplitud

si X es par $\Rightarrow y=0$

si X es impar $\Rightarrow y=1$

$$P(X=\text{impar}) = \sinh(2) e^{-2}$$

$$P(X=\text{par}) = \cosh(2) e^{-2}$$

c) 4.3) $F_X(x) = \frac{12x}{\pi^2(e^x+1)} \mathbb{1}\{x>0\}$

L a) $Y = aX + b$ ($a \neq 0, b \in \mathbb{R}$) $\Rightarrow F_Y(y) =$
 $= P(Y \leq y) = P(aX + b \leq y) = P(X \leq \frac{y-b}{a}) = F_X(\frac{y-b}{a})$
 $\Rightarrow F_Y(y) = F_X(\frac{y-b}{a}) \cdot \frac{1}{a} = \frac{12(\frac{y-b}{a})}{\pi^2(e^{\frac{y-b}{a}}+1)a} \mathbb{1}\{\frac{y-b}{a} > 0\}$
 $\downarrow$
 $\mathbb{1}\{y > b\}$

$$\therefore b) Y = -X^3 \Rightarrow X = \sqrt[3]{-Y} \quad F_Y(y) = F_X(g^{-1}(y))$$

$$\boxed{F_Y(y) = \frac{F_X(\sqrt[3]{-y})}{-3X^2}} = \frac{F_X(\sqrt[3]{-y})}{-3(\sqrt[3]{-y})^2} = \frac{12\sqrt[3]{-y}}{-\pi^2(e^{\frac{1}{\sqrt[3]{-y}}+1})3\sqrt[3]{-y}^2} \quad \mathbb{1}\{y \leq 0\}$$

$$c) Y = X + X^{-1} \Rightarrow YX = X^2 + 1 \Rightarrow X^2 - XY + 1 = 0$$

$$X_1 = \frac{Y + \sqrt{Y^2 - 4}}{2} \quad X_2 = \frac{Y - \sqrt{Y^2 - 4}}{2}$$

$$\boxed{F_Y(y) = \frac{F_X(x)}{1-x^{-2}} \Big|_{x=\frac{y+\sqrt{y^2-4}}{2}} + \frac{F_X(x)}{1-x^{-2}} \Big|_{x=\frac{y-\sqrt{y^2-4}}{2}}}$$

$$\mathbb{1}\{2 < y < \infty\}$$

$$d) Y = X^2 - 3X \rightarrow X^2 - 3X - Y = 0$$

idem inciso anterior

$$4.4) \theta \sim U(-\pi/2, \pi/2) \quad \tan(\theta) = X$$

$$F_\theta(t) = \frac{1}{\pi} \mathbb{1}\{-\pi/2 < t < \pi/2\}$$

$$F_\theta(t) = \left(\frac{t}{\pi} + \frac{1}{2}\right) \mathbb{1}\{-\pi/2 < t < \pi/2\} + \mathbb{1}\{t > \pi/2\}$$

$$F_X(x) = P(\tan \theta \leq x) = P(\theta \leq \tan^{-1}(x)) = F_\theta(\tan^{-1}(x)) = \frac{\tan^{-1}(x)}{\pi} + \frac{1}{2}$$

$$\Rightarrow F_X(x) = \frac{1}{\pi(1+x^2)} \mathbb{1}\{x \in \mathbb{R}\}$$

4.5) $\phi \sim U(-\pi, \pi) \rightarrow F_{\phi}(t) = \frac{1}{2\pi} \mathbb{1}_{\{-\pi < t < \pi\}}$

a) $C = \cos \phi$

$$F_{\phi}(t) = \int_{-\pi}^t \frac{1}{2\pi} du = \frac{t}{2\pi} + \frac{1}{2} \mathbb{1}_{\{-\pi < t < \pi\}} + \mathbb{1}_{\{t > \pi\}}$$

$F_C(c) =$

$= P(C \leq c) = P(\cos \phi \leq c) = P(\phi \leq \cos^{-1}(c)) =$

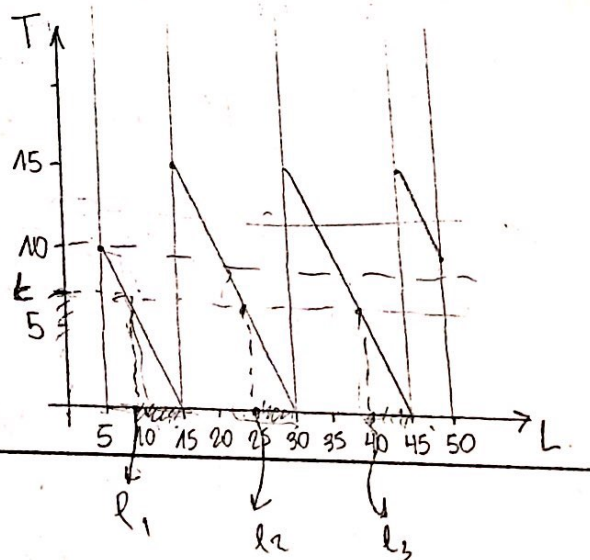
$= F_{\phi}(\cos^{-1}(c)) = \frac{\cos^{-1}(c)}{2\pi} + \frac{1}{2}$

$\Rightarrow F_C(c) = \frac{-1}{2\pi\sqrt{1-c^2}} + \frac{1}{2} \mathbb{1}_{\{-1 < c < 1\}}$

b) $P(|C| < 0.5) = \int_{-0.5}^{0.5} \left(\frac{-1}{2\pi\sqrt{1-c^2}} + \frac{1}{2} \right) dc = \frac{1}{3}$

4.6) $L \sim U(5, 50) \rightarrow F_L(l) = \frac{1}{45} \mathbb{1}_{\{5 < l < 50\}}$
el tren llega 7:15, 7:30, 7:45, 8:00

$$T = \begin{cases} 15-L & \text{si } 5 < l \leq 15 \\ 30-L & \text{si } 15 < l \leq 30 \\ 45-L & \text{si } 30 < l \leq 45 \\ 60-L & \text{si } 45 < l \leq 50 \\ 0 & \text{si } l < 5 \wedge l > 50 \end{cases}$$



$$F_T(t) = P(T \leq t) = \begin{cases} 0 & \text{si } t < 0 \\ 1 & \text{si } t > 15 \\ * & \text{si } t \in (0, 10) \\ ** & \text{si } t \in (10, 15) \end{cases}$$

$$* P(T \leq t) = P(15 - L \leq t) + P(30 - L \leq t) + P(45 - L \leq t) =$$

$$= P(L \geq 15 - t) + P(L \geq 30 - t) + P(L \geq 45 - t) =$$

$$= \int_{15-t}^{15} \frac{1}{45} dl + \int_{30-t}^{30} \frac{1}{45} dl + \int_{45-t}^{45} \frac{1}{45} dl =$$

$$= \frac{15 - (15 - t)}{45} + \frac{30 - (30 - t)}{45} + \frac{45 - (45 - t)}{45} =$$

$$= \frac{t}{45} + \frac{t}{45} + \frac{t}{45} = \frac{t}{15}$$

$$** P(T \leq t) = P(5 < L < 15) + P(30 - L \leq t) + P(45 - L \leq t) + P(60 - L \leq t) =$$

$$= \int_5^{15} \frac{1}{45} dl + \int_{30-t}^{30} \frac{1}{45} dl + \int_{45-t}^{45} \frac{1}{45} dl + \int_{60-t}^{50} \frac{1}{45} dl =$$

4.8

$$P = \frac{10}{45} + \frac{t}{45} + \frac{t}{45} + \frac{t-10}{45} = \frac{t}{15}$$

$$\Rightarrow F_T(t) = \begin{cases} 0 & \text{si } t < 0 \\ \frac{t}{15} & \text{si } 0 \leq t < 15 \\ 1 & \text{si } t \geq 15 \end{cases}$$

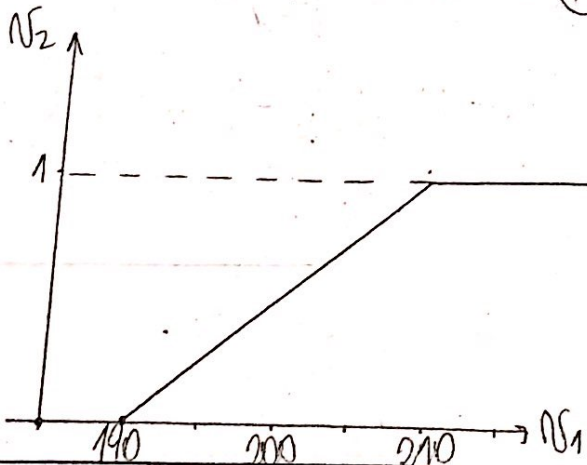
$$4.8 \Rightarrow F_T(t) = \frac{1}{15} \mathbb{1}\{0 < t < 15\}$$

$$4.7) V_1 \sim U(180, 220) \rightarrow F_{V_1}(v_1) = \frac{1}{40} \mathbb{1}\{180 < v_1 < 220\}$$

$$g(v_1) = \frac{v_1 - 190}{20} \mathbb{1}\{190 \leq v_1 \leq 210\} + \mathbb{1}\{v_1 > 210\}$$

$$F_{V_2}(v_2) = P(V_2 \leq v_2) = \begin{cases} 0 & \text{si } v_2 < 0 \\ * & \text{si } 0 \leq v_2 < 1 \\ 1 & \text{si } v_2 \geq 1 \end{cases}$$

** si $v_2 = 0$
*** si $v_2 = 1$



$$* P(V_2 \leq v_2) = P\left(\frac{v_1 - 190}{20} \leq v_2\right)$$

$$= P(v_1 \leq 20v_2 + 190) = \int_{190}^{20v_2 + 190} \frac{1}{40} dv_1 = \frac{20v_2}{40} = \frac{v_2}{2}$$

$$** P(180 < v_1 < 190) = \frac{10}{40}$$

$$*** P(210 < v_1 < 220) = \frac{10}{40}$$

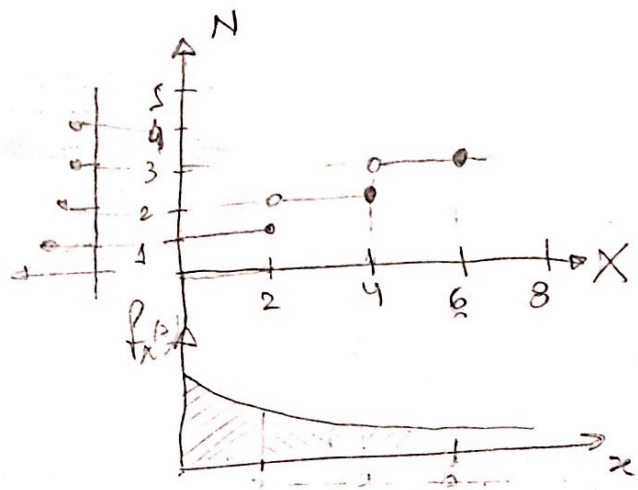
4.8) $X \sim \text{EXP}(1/8)$

N : # trials

$$P(N=n) = P(2(n-1) < X \leq 2n)$$

$$= \int_{2(n-1)}^{2n} \frac{1}{8} e^{-x/8} dx =$$

$$= e^{-\frac{2(n-1)}{8}} - e^{-\frac{2n}{8}} = e^{-\frac{n-1}{4}} - e^{-\frac{n}{4}}, n=1,2,3,\dots$$



4.9) a) $U=X$ $V=X+Y$

$U \backslash V$	-4	-3	-2	-1	0	1	2	3	4
-2	$1/16$	$1/16$	0	$1/8$	$1/8$	0	0	0	0
-1	0	$1/16$	$1/16$	0	$1/8$	$1/8$	0	0	0
1	0	0	0	0	0	0	$1/16$	$1/16$	0
2	0	0	0	0	0	0	0	$1/16$	$1/16$

$$P_{UV}(-2, -4) = P_{XY}(-2, -2) = 1/16 \quad P_{UV}(-2, -3) = P_{XY}(-2, -1) = 1/16$$

$$P_{UV}(-2, -2) = P_{XY}(-2, 0) = 0 \quad P_{UV}(-2, -1) = P_{XY}(-2, 1) = 1/8$$

$$P_{UV}(-2, 0) = P_{XY}(-2, 2) = 1/8 \quad P_{UV}(-1, -4) = 0$$

b) $U = \min(X, Y)$ $V = \max(X, Y)$

$U \backslash V$	-2	-1	1	2
-2	$\frac{1}{16}$	$\frac{2}{16}$	$\frac{1}{8}$	$\frac{1}{8}$
-1	$\frac{2}{16}$	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{8}$
1	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{2}{16}$
2	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{2}{16}$	$\frac{1}{16}$

$$p_{UV}(-2, -2) = p_{XY}(-2, -2) = \frac{1}{16}$$

$$p_{UV}(-2, -1) = p_{XY}(-2, -1) + p_{XY}(-1, -2)$$

c) $U = X^2 + Y^2$ $V = \frac{Y}{X}$

$U \backslash V$	-1	$-\frac{1}{2}$	$\frac{1}{2}$	1
2	$\frac{1}{8}$	0	0	$\frac{2}{16}$
5	0	$\frac{1}{8}$	$\frac{2}{16}$	0
8	$\frac{1}{8}$	0	0	$\frac{2}{16}$

$$p_{UV}(2, -1) = p_{XY}(1, -1) + p_{XY}(-1, 1)$$

$$p_{UV}(2, -\frac{1}{2}) = 0$$

$U \backslash V$	-2	2
2	0	0
5	$\frac{1}{8}$	$\frac{2}{16}$
8	0	0

4.10) a) $(U, V) = A(X, Y)^T + B$

$$(U, V) = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix} + \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$$

$$\begin{cases} U = a_{11}X + a_{12}Y + b_1 \\ V = a_{21}X + a_{22}Y + b_2 \end{cases} \Rightarrow \begin{matrix} Y = \frac{V - \frac{a_{21}U + a_{21}b_1 - b_2}{a_{11}}}{a_{22} - \frac{a_{21}a_{12}}{a_{11}}} \\ * \end{matrix}$$

$$F_{UV}(u, v) = \frac{F_{XY}(x(u, v), y(u, v))}{\left| \frac{\partial(U, V)}{\partial(X, Y)} \right|}$$

$$* X = \frac{U - b_1 - a_{12} \left(\frac{V - \frac{a_{21}U + a_{21}b_1 - b_2}{a_{11}}}{a_{22} - \frac{a_{21}a_{12}}{a_{11}}} \right)}{a_{11}}$$

$$\Rightarrow \left| \frac{\partial(U, V)}{\partial(X, Y)} \right| = \left| \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \right| = a_{11}a_{22} - a_{12}a_{21} \neq 0 \quad \text{for ser invertible}$$

b) $(U,V) = (\min(X,Y), \max(X,Y))$

$$(U,V) = \begin{cases} (X,Y), & X < Y \quad ① \\ (Y,X), & X \geq Y \quad ② \end{cases}$$

$$F_{UV}(u,v) = \frac{F_{XY}(x(u,v), y(u,v))}{|J_1|} + \frac{F_{XY}(x(u,v), y(u,v))}{|J_2|}$$

$$J_1 = \left| \frac{\partial(U,V)}{\partial(X,Y)} \right| = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1 \quad J_2 = \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = -1$$

$$\Rightarrow F_{UV}(u,v) = 2 F_{XY}(x(u,v), y(u,v))$$

c) $(U,V) = (X^2 + Y^2, Y/X)$

$U = X^2 + Y^2 \Rightarrow \begin{matrix} (-X, Y) \\ (X, -Y) \end{matrix} > (U,V) \text{ NO ES UNO A UNO}$

$$\left. \begin{aligned} ① U &= (-X)^2 + Y^2 \\ V &= Y/(-X) \end{aligned} \right\} \begin{aligned} |J_1| &= \left| \frac{\partial(U,V)}{\partial(X,Y)} \right| = \begin{vmatrix} 2X & 2Y \\ Y/X^2 & -1/X \end{vmatrix} = -2 - 2V^2 \\ ② U &= X^2 + (-Y)^2 \\ V &= -Y/X \end{aligned} \left\{ \begin{aligned} |J_2| &= \begin{vmatrix} 2X & 2Y \\ Y/X^2 & -1/X \end{vmatrix} = -2 - 2V^2 \end{aligned} \right.$$

$$\Rightarrow F_{UV}(u,v) = \frac{F_{XY}(x(u,v), y(u,v))}{2 + 2v^2} + \frac{F_{XY}(x(u,v), y(u,v))}{2 + 2v^2}$$

4.11) $Z_1 \sim N(0,1)$ independientes
 $Z_2 \sim N(0,1)$

a) 1) $U = Z_1 + Z_2$
 $V = Z_1 - Z_2 \Rightarrow \left| \frac{\partial(U,V)}{\partial(Z_1,Z_2)} \right| = \left| \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \right| = 2$

$$F_{Z_1 Z_2}(z_1, z_2) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z_1^2}{2}} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{z_2^2}{2}} = \frac{1}{2\pi} e^{-\frac{(z_1^2 + z_2^2)}{2}} \mathbb{1}_{\{z_1, z_2 \in \mathbb{R}\}}$$

$$F_{UV}(u, v) = \frac{1}{2\pi} \frac{e^{-\left(\frac{(u+v)^2}{2} + \frac{(u-v)^2}{2}\right)}}{2} = \frac{1}{4\pi} e^{-\frac{(u^2 + v^2)}{2}} \mathbb{1}_{\{u, v \in \mathbb{R}\}}$$

II) $(U, V) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} Z_1 \\ Z_2 \end{pmatrix} \quad \theta \in (0, 2\pi)$

$$U = Z_1 \cos \theta - Z_2 \sin \theta \Rightarrow U^2 + V^2 = Z_1^2 + Z_2^2$$

$$V = Z_1 \sin \theta + Z_2 \cos \theta$$

$$F_{UV}(u, v) = \frac{1}{2\pi} e^{-\frac{(u^2 + v^2)}{2}} \times 1 \mathbb{1}_{\{u, v \in \mathbb{R}\}}$$

$$\left| \frac{\partial(U, V)}{\partial(Z_1, Z_2)} \right| = \left| \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \right| = 1$$

$$* U = (-z_1)^2 + z_2^2$$

$$V = z_2 / (-z_1)$$

$$|J_1| = 2 + 2V^2$$

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$$** U = z_1^2 + (-z_2)^2$$

$$V = -z_2 / z_1$$

$$|J_2| = 2 + 2V^2$$

$$III) U = z_1^2 + z_2^2 \Rightarrow (z_1, -z_2) > (u, v) \text{ NO ES UNO A UNO}$$

$$V = \frac{z_2}{z_1}$$

$$* \Rightarrow F_{UV}(u, v) = \frac{2F_{XY}(x(u, v), y(u, v))}{2 + 2v^2} = \frac{1}{\pi(2 + 2v^2)} e^{-\frac{u}{2}} \mathbb{1}_{\{u > 0, v \in \mathbb{R}\}}$$

$$b) I) F_{UV}(u, v) = \frac{1}{4\pi} e^{-\frac{(\frac{u^2}{2} + \frac{v^2}{2})}{2}} = \frac{1}{4\pi} e^{-\frac{u^2}{4}} \times \frac{1}{2\sqrt{\pi}} e^{-\frac{v^2}{4}}$$

$$\underbrace{\frac{1}{2\sqrt{\pi}} e^{-\frac{u^2}{4}}}_{F_U(u)} \underbrace{\frac{1}{2\sqrt{\pi}} e^{-\frac{v^2}{4}}}_{F_V(v)}$$

$\Rightarrow$ son independientes

$$II) F_{UV}(u, v) = \frac{1}{2\pi} e^{-\frac{(u^2 + v^2)}{2}} = \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} \times \frac{1}{\sqrt{2\pi}} e^{-\frac{v^2}{2}}$$

$$\underbrace{\frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}}}_{F_U(u)} \underbrace{\frac{1}{\sqrt{2\pi}} e^{-\frac{v^2}{2}}}_{F_V(v)}$$

$\Rightarrow$ son independientes

$$III) F_{UV}(u, v) = \frac{1}{\pi} e^{-\frac{u}{2}} = \frac{1}{\pi(1 + v^2)} \times \frac{1}{2} e^{-\frac{u}{2}}$$

$$\underbrace{\frac{1}{\pi(1 + v^2)}}_{F_V(v)} \underbrace{\frac{1}{2} e^{-\frac{u}{2}}}_{F_U(u)}$$

$\Rightarrow$ son independientes

2}

$$P(z_1^2 + z_2^2 > 4) \stackrel{(\text{III})}{=} P(U > 4) = \int_4^{+\infty} \frac{1}{2} e^{-\frac{u}{2}} du = e^{-\frac{4}{2}} = e^{-2} = 0,135$$

$$P(z_2 > \sqrt{3} z_1) = P(z_2/z_1 > \sqrt{3}) \stackrel{(\text{III})}{=} P(V > \sqrt{3}) = \int_{\sqrt{3}}^{+\infty} \frac{1}{\pi(1+v^2)} dv = 0,167$$

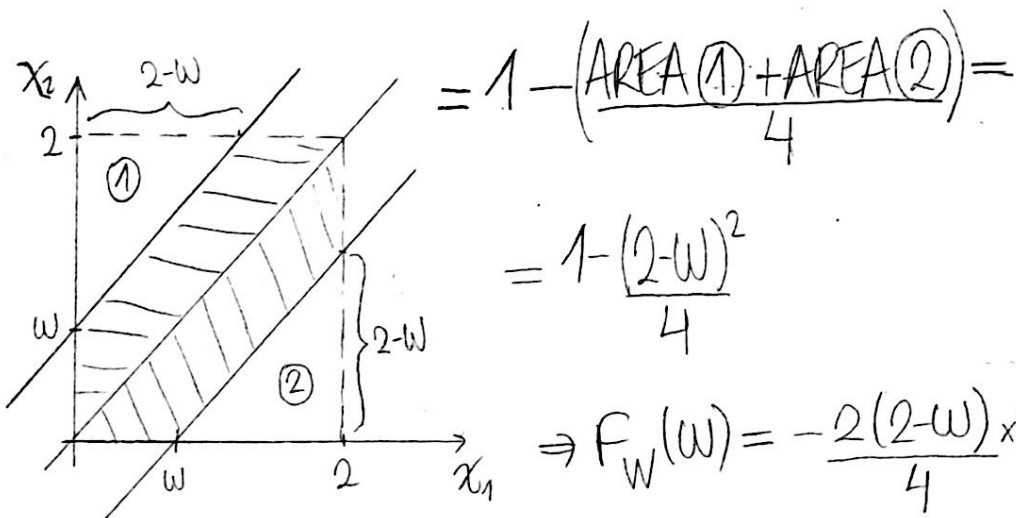
4.12) $X_1 \sim U(0,2)$ $\Rightarrow$ independentes $U = \min(X_1, X_2)$
 $X_2 \sim U(0,2)$ $F_{X_1 X_2}(x_1, x_2) = \frac{1}{4}$ $V = \max(X_1, X_2)$

a)

$$F_{UV}(u, v) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2} \quad \text{for } \{0 < u < v < 2\}$$

b) $W = V - U$

$$P(W \leq w) = P(X_1 - X_2 \leq w, X_1 > X_2) + P(X_2 - X_1 \leq w, X_1 \leq X_2)$$



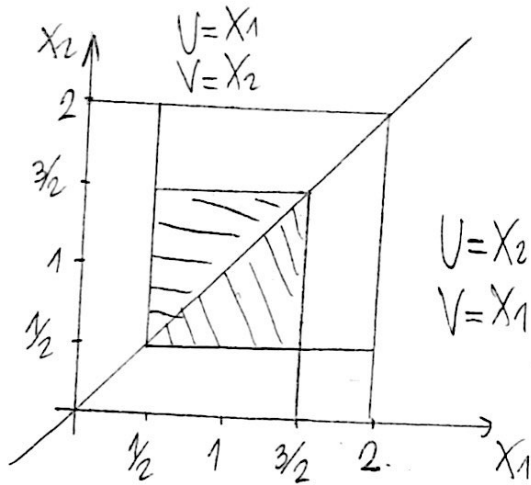
$$= 1 - \frac{(\text{AREA (1)} + \text{AREA (2)})}{4} =$$

$$= 1 - \frac{(2-w)^2}{4}$$

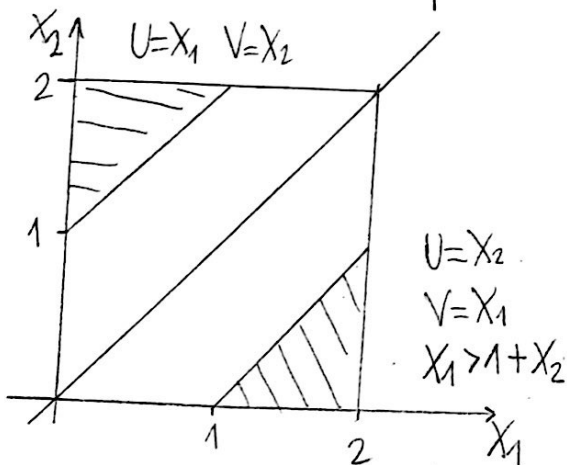
$$\Rightarrow F_W(w) = -\frac{2(2-w)}{4} \times (-1) =$$

$$= \frac{2-w}{2} = 1 - \frac{w}{2} \quad \text{for } \{0 < w < 2\}$$

$$c) P(U > \frac{1}{2}, V < \frac{3}{2}) = \frac{1}{4}$$

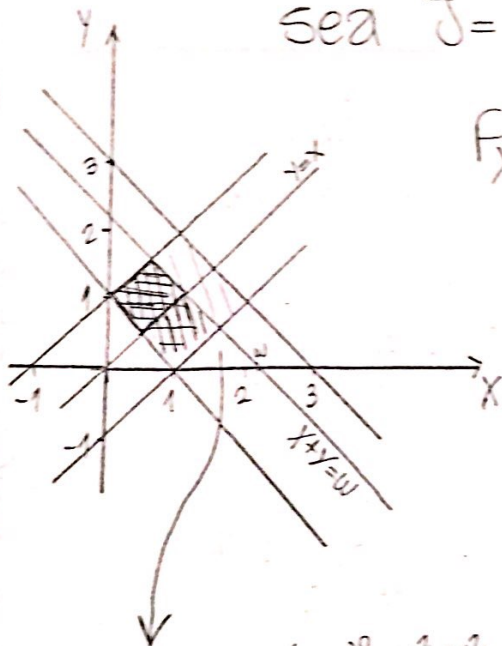


$$P(V > 1+U) = \frac{1}{4}$$



4.13) $(X, Y) \sim U(A)$

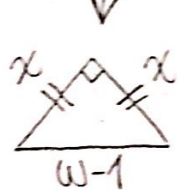
sea $J = \mathbb{1}\{X < Y\} \rightarrow P(J=1) = \frac{1}{2}$



$$F_{XY}(x, y) = \frac{1}{2}$$

$$P(X+Y \leq w) = P(Y \leq w-X) = \sqrt{2} \times \frac{w-1}{2}$$

$$P(X+Y \leq w, J=1) = \frac{P(X+Y \leq w)}{2} = \frac{\sqrt{2} \times (w-1)}{4}$$



$$(w-1)^2 = x^2 + x^2$$

$$(w-1)^2 = 2x^2$$

$$\frac{(w-1)^2}{2} = x^2$$

$$\frac{w-1}{\sqrt{2}} = x$$

$\Rightarrow$ son independientes

4.14) $X_1 \sim \text{EXP}(\lambda_1)$ $X_2 \sim \text{EXP}(\lambda_2)$
 $U = \min(X_1, X_2)$ $V = \max(X_1, X_2)$ $W = V - U$
 $J = \mathbb{1}\{U = X_1\} + 2\mathbb{1}\{U = X_2\}$

a) $U \sim \text{EXP}(\lambda_1 + \lambda_2)$
 $F_U(u) = (\lambda_1 + \lambda_2) e^{-(\lambda_1 + \lambda_2)u} \mathbb{1}\{u > 0\}$

$$b) p_J(1) = P(J=1) = P(X_1 < X_2) = \frac{\lambda_1}{\lambda_1 + \lambda_2}$$

$$p_J(2) = P(J=2) = P(X_1 > X_2) = \frac{\lambda_2}{\lambda_1 + \lambda_2}$$

$$\begin{aligned} c) F_W(w) &= F_{W|J=1}(w) \times P(J=1) + F_{W|J=2}(w) \times P(J=2) \\ &= \lambda_2 e^{-\lambda_2 w} \frac{\lambda_1}{\lambda_1 + \lambda_2} + \lambda_1 e^{-\lambda_1 w} \frac{\lambda_2}{\lambda_1 + \lambda_2} \mathbb{1}\{w > 0\} \end{aligned}$$

$$\begin{aligned} d) \text{ probar } P(J=1, U \leq u) &= P(J=1) \times P(U \leq u) \\ &\Rightarrow U \text{ y } J \text{ son independientes} \end{aligned}$$

$$\begin{aligned} e) \text{ probar } F_{WU}(w, u) &= F_W(w) \times F_U(u) \\ &\Rightarrow U \text{ y } W \text{ son independientes} \end{aligned}$$

4.15) $X \sim \text{EXP} \sim (5)$: "tempo entre llamadas de Juan"
 $Y \sim \text{EXP} \sim (10)$: " " " " " Pedro"

5 - 60 min

$$X - 1 \text{ min} \Rightarrow \frac{1}{12}$$

10 - 60 min

$$X - 1 \text{ min} \Rightarrow \frac{1}{6}$$

$$F_X(x) = \frac{1}{12} e^{-\frac{x}{12}} \mathbb{1}_{\{x > 0\}}$$

$$F_Y(y) = \frac{1}{6} e^{-\frac{y}{6}} \mathbb{1}_{\{y > 0\}}$$

$$W = \max(X, Y) - \min(X, Y)$$

$$f_W(w) = \frac{\lambda_1}{\lambda_1 + \lambda_2} \lambda_2 e^{-\lambda_2 w} f_{Y|X} + \frac{\lambda_2}{\lambda_1 + \lambda_2} \lambda_1 e^{-\lambda_1 w} f_{X|Y}$$

denom: $P(X < Y)$ $P(Y < X)$

$$a) P(U \leq 5) = 1 - e^{-(\frac{1}{12} + \frac{1}{6})5} = 0,713$$

$\uparrow$
 $U \sim \text{EXP}(\lambda_1 + \lambda_2)$

$$b) P(X < Y) = \frac{1/12}{1/12 + 1/6} = \frac{1}{3}$$

$$c) P(U \leq 5 | X < Y) = P(U \leq 5) = 0,713$$

$\uparrow$
 U y J son independientes

$$d) P(X < Y | U > 5) = P(X < Y) = \frac{1}{3}$$

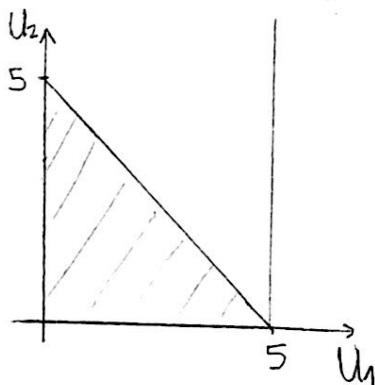
$$e) P(\overbrace{U_2 - U_1}^W > 5) = P(U_2 > 5) = 1 - P(U \leq 5) =$$

$$= 1 - 0,713 = 0,287$$

$$f) P(U < 5) = 0,713$$

$$g) \underbrace{P(U_2 < 5 | U_1 > 5)}_{0,713} \times \underbrace{P(U_1 > 5)}_{0,287} + \underbrace{P(U_1 + U_2 < 5 | U_1 < 5)}_{\underbrace{P(U_1 < 5)}_{0,713}}$$

$$P(U_1 + U_2 < 5 | U_1 < 5) = \frac{P(U_1 + U_2 < 5 \wedge U_1 < 5)}{P(U_1 < 5)}$$



$$= \frac{\int_0^5 \int_0^{5-u_2} \frac{1}{4} e^{-\frac{u_1}{4}} \times \frac{1}{4} e^{-\frac{u_2}{4}} du_1 du_2}{0,713} =$$

$$= \frac{0,355}{0,713} = 0,498$$

$$\Rightarrow 0,713 \times 0,287 + 0,498 \times 0,713 = 0,6$$

4.16) $X \sim \text{EXP}(\lambda)$

$Y \sim \text{EXP}(\lambda)$

$U = X + Y$

$V = \frac{X}{X+Y}$

a) a)

$$\left| \frac{\partial(U,V)}{\partial(X,Y)} \right| = \left| \begin{pmatrix} 1 & 1 \\ \frac{(X+Y)-X}{(X+Y)^2} & \frac{-X}{(X+Y)^2} \end{pmatrix} \right| = \frac{-X}{(X+Y)^2} - \frac{(X+Y)-X}{(X+Y)^2} = \frac{-(X+Y)}{(X+Y)^2} =$$

$$= \frac{-1}{X+Y} = \frac{-1}{U}$$

$$\Rightarrow F_{UV}(u, r) = \frac{\lambda^2 e^{-\lambda(x+y)}}{1/u} = \lambda^2 e^{-\lambda u} u \mathbb{1}_{\{u \in \mathbb{R}, |r| \leq 1\}}$$

b) $U \sim \text{GAMMA}(2, \lambda)$

$F_U(u) = \lambda^2 u e^{-\lambda u} = F_{UV}(u, r)$

$\Rightarrow V \sim U(1, 0)$

$\Rightarrow$ son independientes

$$4.17) \begin{matrix} X \sim N(0,1) \\ Y \sim N(0,1) \end{matrix} \Rightarrow F_{XY}(x,y) = \frac{1}{2\pi} e^{-\frac{(x^2+y^2)}{2}} \mathbb{1}_{\{x,y \in \mathbb{R}\}}$$

$$\begin{matrix} X = r \cos \theta \\ Y = r \sin \theta \end{matrix} \Rightarrow \left| \frac{\partial(X,Y)}{\partial(r,\theta)} \right| = r$$

$$\Rightarrow F_{R\theta}(r,\alpha) = \frac{1}{2\pi} e^{-\frac{r^2}{2}} r \mathbb{1}_{\{r>0, \alpha \in (0, 2\pi)\}}$$

$$F_R(r) = \int_0^{2\pi} \frac{1}{2\pi} e^{-\frac{r^2}{2}} r d\theta = e^{-\frac{r^2}{2}} r \mathbb{1}_{\{r>0\}}$$

$$\Rightarrow F_\theta(\alpha) = \frac{1}{2\pi} \mathbb{1}_{\{\alpha \in (0, 2\pi)\}}$$

$$4.18) \begin{matrix} U_1 \sim U(0,1) \\ U_2 \sim U(0,1) \end{matrix} \quad \begin{matrix} (Z_1, Z_2) = (R \cos \theta, R \sin \theta) \\ R = \sqrt{-2 \ln(U_1)}, \quad \theta = 2\pi U_2 \end{matrix}$$

$$\begin{aligned} a) \quad F_R(r) &= P(R \leq r) = P(\sqrt{-2 \ln(U_1)} \leq r) = P(U_1 \geq e^{-\frac{r^2}{2}}) = \\ &= 1 - P(U_1 \leq e^{-\frac{r^2}{2}}) = 1 - F_{U_1}(e^{-\frac{r^2}{2}}) = 1 - e^{-\frac{r^2}{2}} \end{aligned}$$

$$\Rightarrow F_R(r) = r e^{-\frac{r^2}{2}} \mathbb{1}_{\{r>0\}}$$

$$F_\theta(\psi) = P(\theta \leq \psi) = P(2\pi U_2 \leq \psi) = P(U_2 \leq \frac{\psi}{2\pi}) = \frac{\psi}{2\pi}$$

$$\Rightarrow F_\theta(\psi) = \frac{1}{2\pi} \mathbb{1}_{\{\psi \in (0, 2\pi)\}}$$

4.24

$$b) F_{Z_1 Z_2}(z_1, z_2) = F_{R\theta}(r, \varphi) = \frac{1}{2\pi} \frac{r e^{-\frac{r^2}{2}}}{\left| \frac{\partial(z_1, z_2)}{\partial(R, \theta)} \right|} =$$

a)

$$= \frac{1}{2\pi} e^{-\frac{r^2}{2}} = \frac{1}{2\pi} e^{-\frac{(z_1^2 + z_2^2)}{2}} \mathbb{1}_{\{z_1, z_2 > 0\}}$$

$$c) F_{Z_1 Z_2}(z_1, z_2) = \underbrace{\frac{1}{\sqrt{2\pi}} e^{-\frac{z_1^2}{2}}}_{Z_1 \sim N(0,1)} \times \underbrace{\frac{1}{\sqrt{2\pi}} e^{-\frac{z_2^2}{2}}}_{Z_2 \sim N(0,1)} \Rightarrow \text{son independientes}$$

*

4.19) X, Y independientes

$$P(X=1) = P(X=2) = \dots = P(X=36) = \frac{1}{36}$$

$$P(Y=1) = P(Y=2) = P(Y=3) = P(Y=4) = \frac{1}{4}$$

hallar $p_{X+Y}(x+y)$

$y \backslash x$	1	2	3	...	35	36
1	2	3	4		36	37
2	3	4	5		37	38
3	4	5	6		38	39
4	5	6	7		39	40

 $p_{X+Y}(40)$

$$p_{X+Y}(2) = P(X=1, Y=1) = \frac{1}{4} \times \frac{1}{36} =$$

$$p_{X+Y}(3) = P(X=1, Y=2) + P(X=2, Y=1) = 2 \left(\frac{1}{4} \times \frac{1}{36} \right) = p_{X+Y}(39)$$

$$p_{X+Y}(4) = 3 \left(\frac{1}{4} \times \frac{1}{36} \right) = p_{X+Y}(38)$$

$$p_{X+Y}(5) = p_{X+Y}(6) = \dots = p_{X+Y}(37) = 4 \left(\frac{1}{4} \times \frac{1}{36} \right)$$

4.20) $X \sim U$
 $L \sim \text{Poisson}(2)$
 $M \sim \text{Poisson}(8)$ independientes

$$U = \frac{X}{2}$$



a) $L+M \sim \text{Poisson}(10)$

$$p_{L+M}(n) = \frac{10^n e^{-10}}{n!}, n \geq 0$$

b) $M | L+M=10 \sim \text{BINOMIAL}(10, \frac{8}{10})$

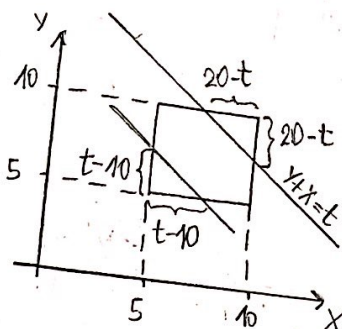
$$p_{M|L+M=10}(m) = \binom{10}{m} \left(\frac{8}{10}\right)^m \left(\frac{2}{10}\right)^{10-m}, m \geq 0$$

si $\frac{5}{2}$
 $P(L$
 $\ast$ si

c) $P(M > 2 | L+M=10) = 1 - P(M \leq 2 | L+M=10) =$
 $= 1 - \left(\binom{10}{0} \left(\frac{8}{10}\right)^0 \left(\frac{2}{10}\right)^{10} + \binom{10}{1} \left(\frac{8}{10}\right)^1 \left(\frac{2}{10}\right)^9 + \binom{10}{2} \left(\frac{8}{10}\right)^2 \left(\frac{2}{10}\right)^8 \right) =$
 $= 1$

4.21) $X \sim U(5, 10)$
 $Y \sim U(5, 10)$

$\ast$ a) $T = X+Y$



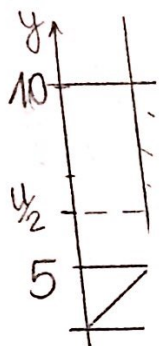
$$P(T \leq t) = P(X+Y \leq t) = \begin{cases} 0 & \text{si } t < 10 \\ 1 & \text{si } t \geq 20 \\ \ast & \text{si } 10 \leq t < 15 \\ \ast\ast & \text{si } 15 \leq t < 20 \end{cases}$$

$$\ast \frac{(t-10)^2}{2} \times \frac{1}{25} = \frac{(t-10)^2}{50}$$

$$\ast\ast 1 - \frac{(20-t)^2}{2} \times \frac{1}{25} = 1 - \frac{(20-t)^2}{50}$$

$$1) X \sim U(5, 10)$$

$$U = \frac{X}{2}$$

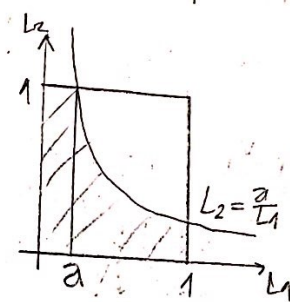
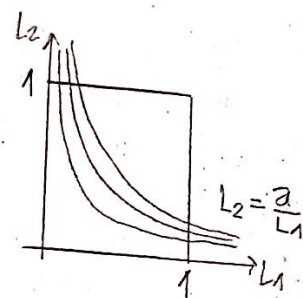


$$\Rightarrow F_{X+Y}(t) = \frac{t-10}{25} \mathbb{1}_{\{10 \leq t < 15\}} + \frac{20-t}{25} \mathbb{1}_{\{15 \leq t < 20\}}$$

$$b) P(X+Y < 16) = \int_{10}^{15} \frac{t-10}{25} dt + \int_{15}^{16} \frac{20-t}{25} dt = 0,68$$

4.22) $L_1 \sim U(0, 1)$
 $L_2 \sim U(0, 1)$
independientes

$$A = L_1 \times L_2$$



$$a) P(A \leq a) = P(L_1 \times L_2 \leq a) =$$

$$= \begin{cases} 1 & \text{si } a \geq 1 \\ 0 & \text{si } a < 0 \\ * & \text{si } 0 \leq a < 1 \end{cases}$$

$$P(* a + \int_a^1 \int_0^{a/L_1} dL_2 dL_1 = a + \int_a^1 \frac{a}{L_1} dL_1 = a + a \ln\left(\frac{1}{a}\right) = a - a \ln(a)$$

$$P(\Rightarrow F_A(a) = 1 - (\ln(a) + 1) = -\ln(a) \mathbb{1}_{\{0 \leq a < 1\}}$$

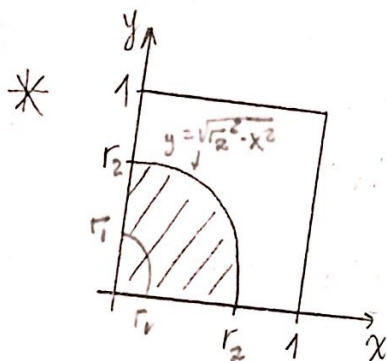
$$F_U' b) P(A > 1/4) = \int_{1/4}^1 -\ln(a) da = 0,403$$

4.21

$$4.23) X \sim U(0,1) \\ Y \sim U(0,1)$$

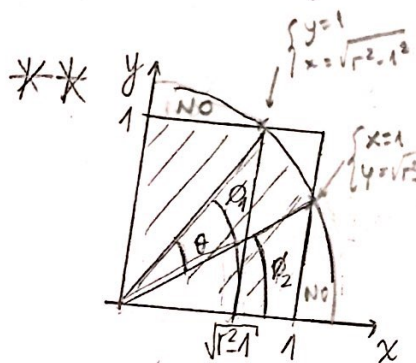
$$R = \sqrt{X^2 + Y^2}$$

$$a) a) P(R \leq r) = P(\sqrt{X^2 + Y^2} \leq r) = \begin{cases} 1 & \text{si } r \geq \sqrt{2} \\ 0 & \text{si } r < 0 \\ * & \text{si } 0 \leq r < 1 \\ ** & \text{si } 1 \leq r < \sqrt{2} \end{cases}$$



$$P(R \leq r) = \int_0^r \int_0^{\sqrt{r^2 - x^2}} dy dx = \frac{\pi r^2}{4} \quad \text{for } r \leq 1$$

(4^{to} de círculo)



$$P(R \leq r) = \sqrt{r^2 - 1} + \int_0^{\sqrt{r^2 - 1}} \int_0^{\sqrt{r^2 - x^2}} dy dx =$$

formar un $\square$
 $A(\square) = \sqrt{r^2 - 1}$

$$= A(\square) + A(\triangle) + A(\text{sector}) =$$

$$= \frac{\sqrt{r^2 - 1}}{2} + \frac{\sqrt{r^2 - 1}}{2} + \pi r^2 \frac{\theta}{2\pi}, \quad \theta = \phi_1 - \phi_2$$

densidad

$$\Rightarrow F_R(r) = \frac{\pi r^2}{2} \mathbb{1}_{\{0 \leq r < 1\}} + \left(\frac{\pi}{2} \right) \mathbb{1}_{\{1 \leq r < \sqrt{2}\}}$$

$$b) P(R > 1/2) = 1 - P(R \leq 1/2) = 1 - \frac{\pi (1/2)^2}{4} = 0,804$$

*

$$\phi_1 = \arcsin\left(\frac{1}{r}\right) + \arcsin\left(\frac{\sqrt{r^2 - 1}}{r}\right)$$

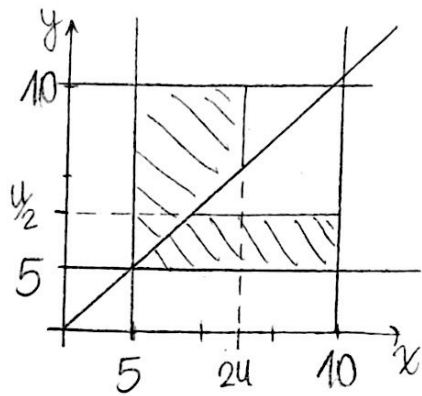
$$\phi_2 = \arccos\left(\frac{1}{r}\right) - \arcsin\left(\frac{\sqrt{r^2 - 1}}{r}\right)$$

$$\Rightarrow P(R \leq r) = \sqrt{r^2 - 1} + \frac{r^2}{2} (\phi_2 - \phi_1)$$

derivar esto Δ recordar que $[f(u)]^{-1} = \frac{u'}{1+u^2}$

$$4.24) \quad X \sim U(5, 10) \\ Y \sim U(5, 10)$$

$$a) \quad U = \frac{X}{2} \mathbb{1}\{X \leq Y\} + 2Y \mathbb{1}\{X > Y\}$$



$$P(U \leq u) = P\left(\frac{X}{2} \leq u, X \leq Y\right) + P(2Y \leq u, X > Y) = \\ = P(X \leq 2u, X \leq Y) + P(Y \leq \frac{u}{2}, X > Y)$$

$$\Rightarrow F_U(u) = \begin{cases} 0 & \text{si } u \leq \frac{5}{2} \\ 1 & \text{si } u \geq 20 \\ * & \text{si } \frac{5}{2} < u < 5 \\ ** & \text{si } 5 \leq u < 10 \\ *** & \text{si } 10 \leq u < 20 \end{cases}$$

$$* \text{ si } \frac{5}{2} < u < 5$$

$$P(U \leq u) = 1 - \left(\frac{(10-2u)^2}{2} + \frac{25}{2} \right) \times \frac{1}{25} = \frac{1 - (10-2u)^2}{50}$$

$$** \text{ si } 5 \leq u < 10$$

$$P(U \leq u) = \frac{1}{2}$$

$$*** \text{ si } 10 \leq u < 20$$

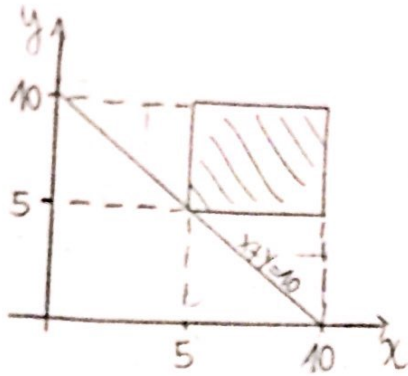
$$P(U \leq u) = 1 - \left(\frac{(10-\frac{u}{2})^2}{2} \right) \times \frac{1}{25} = \frac{1 - (10-\frac{u}{2})^2}{50}$$

deriva de $F_U(u)$

$$F_U(u) = \frac{2(10-2u)}{25} \mathbb{1}\left\{\frac{5}{2} \leq u < 5\right\} + \frac{(10-\frac{u}{2})}{50} \mathbb{1}\{10 \leq u < 20\}$$

$$b) V = \mathbb{1}\{X+Y \leq 10\}$$

$$p_V(1) = P(V=1) = P(X+Y \leq 10) = 0$$



$$p_V(0) = P(V=0) = P(X+Y > 10) = 1$$

c)